Distribution functions

1. In an earlier class activity, you thought about gaps between e-mail messages that I receive. You modeled the length of a gap (measured in hours) with a random variable X having density function f defined by $f(x)=0$, if $x<0$ and $f(x)=5e^{-5x}$, if $x\geq 0$.

a. Find a formula for the distribution function F defined by $F(x)=P(X\leq x)$.

- If $x<0$, what is $F(x)$?

- If $x\geq 0$, you can calculate $F(x)$ as P¿). This probability is given by the integral $\int_0^x f(t)\,dt$. Evaluate this integral to find a formula for $F(x)$.

b. Check your work:

- For this random variable, you should have $F(0)=0$. Does your formula give that?

- You should have $\lim_{x\to\infty} F(x)=1$. Does your formula have the right limiting value?

- Sketch the graph of F. (You may use a graphing calculator or a tool such as Desmos to get the graph. Include $x<0$ in your graph.)

c. The model for a gap between my e-mail messages uses a random variable X with $X \ \ \text{exponential}(5)$ According to the formulas from Durrett, what are E[X] and Var(X)?

E[X] =

Var(X) =

d. According to this model, what is the median gap between e-mail messages?

2. We can think about the distribution function for any random variable, not just for continuous random variables. Suppose that N is a random variable counting the number of heads in three independent coin flips.

a. What distribution does N have?

N ~

b. The distribution function F for N is defined by F(x) = P(N ≤ x). Sketch the graph of this function F, including $x<0$ in your graph.